JAVASCRIPT ASSIGNMENT : 03

Module 3: Variables and Data Types – Multiple Assignment Questions

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Section 1: var let const Based Questions

1. Demonstrate difference between var let and const

Write a program that declares variables using var let and const and try to redeclare and reassign them. Observe output and explain difference.
Hint: Try redeclaring same variable name in same scope.

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2. Create block scope demonstration

Write program using if block and declare variable inside using var and let. Print it outside block and observe behavior.
Hint: Compare accessibility outside block.

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3. Demonstrate function scope using var

Create function and declare variable inside it using var. Try accessing it outside function.
Hint: Understand function scope concept.

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4. Create example showing const behavior with object

Create object using const and try modifying its properties.
Hint: Reassignment is not allowed but property modification is allowed.

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5. Create loop example showing var vs let difference

Create for loop using var and another using let and observe variable value outside loop.
Hint: Check variable availability after loop ends.

Section 2: Data Types Practical Questions

6. Create program to demonstrate all primitive data types

Declare variables of string number boolean null and undefined and print their type using typeof.
Hint: Observe typeof null result carefully.

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7. Create example showing difference between null and undefined

Write program where one variable is declared but not assigned and another is assigned null. Compare output.
Hint: Print both values and their types.

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8. Create object and array example

Create object storing student details and array storing marks. Print values using console.
Hint: Access object properties using dot notation.

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9. Create nested object example

Create object inside object and print nested property value.
Hint: Access nested properties step by step.

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10. Create array with mixed data types

Create array containing string number boolean and object. Print typeof each element.
Hint: Use loop to iterate.

Section 3: typeof Operator Based Questions

11. Write program to check type of user input

Take user input using prompt and check its data type.
Hint: Input from prompt is always string.

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12. Demonstrate typeof for different values

Print typeof for string number boolean null undefined array and object.
Hint: Observe array result carefully.

13. Create program that checks if variable is array

Write logic to check whether variable is array or not.
Hint: typeof will not directly give array.

Section 4: Scope Based Interview Questions

14. Create example of global scope

Declare variable globally and access inside function.
Hint: Global variables are accessible everywhere.

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15. Create shadowing example

Create global variable and inside function declare same variable name using let. Print both.
Hint: Understand variable shadowing.

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16. Create nested block scope example

Declare variable inside nested blocks and test accessibility outside each block.
Hint: Block scope applies to let and const.

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17. Predict output type questions

Write program with mixed data types and ask what will be typeof result.
Hint: Include null and array cases.

Section 5: Tricky Interview Thinking Questions

18. What happens if you declare variable without var let const

Write example and observe behavior.
Hint: It becomes global variable.

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19. Can const object be modified

Create example showing property update and attempt reassignment.
Hint: Only reference cannot change.

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20. Create program showing dynamic typing in JavaScript

Assign number to variable then assign string to same variable and print typeof each time.
Hint: JavaScript is dynamically typed language.

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